

# INTELLIGENT SYSTEMS & ROBOTICS

## ASSIGNMENT 2

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[shreenidhi.bharadwaj@northwestern.edu](mailto:shreenidhi.bharadwaj@northwestern.edu) | [ChristopherFiore2015@u.northwestern.edu](mailto:ChristopherFiore2015@u.northwestern.edu)

### Submissions (1 Submission/Team)

- Presentation Deck and Python Code in a zip file
- Question 1 through Question 4 references chapters within the book Sutton & Barto.

### Question 1: (Chapter 1: Exercise 1.1: Self-Play) {10 Points}

Suppose, instead of playing against a random opponent, the reinforcement learning algorithm described above played against itself, with both sides learning. What do you think would happen in this case? Would it learn a different policy for selecting moves?

### Question 2: (Chapter 1: Exercise 1.2: Symmetries) {10 Points}

Many tic-tac-toe positions appear different but are really the same because of symmetries. How might we amend the learning process described above to take advantage of this? In what ways would this change improve the learning process? Now think again. Suppose the opponent did not take advantage of symmetries. In that case, should we? Is it true, then, that symmetrically equivalent positions should necessarily have the same value?

### Question 3: (Chapter 3: Exercise 3.1) {10 Points}

Devise three example tasks of your own that fit into the MDP framework, identifying for each its states, actions, and rewards. Make the three examples as different from each other as possible. The framework is abstract and flexible and can be applied in many ways. Stretch its limits in some way in at least one of your examples.

### Question 4: (Chapter 3: Exercise 3.3) {10 Points}

Consider the problem of driving. You could define the actions in terms of the accelerator, steering wheel, and brake, that is, where your body meets the machine. Or you could define them farther out - say, where the rubber meets the road, considering your actions to be tire torques. Or you could define them farther in - say, where your brain meets your body, the actions being muscle twitches to control your limbs. Or you could go to a really high level and say that your actions are your choices of where to drive. What is the right level, the right place to draw the line between agent and environment? On what basis is one location of the line to be preferred over another? Is there any fundamental reason for preferring one location over another, or is it a free choice?

### Question 5: (Chapter 2) {60 Points}

Identify a use case relating where MABs might be useful and implement the algorithm to address it. References to use cases provide below.

OR

Consider a multi-arm bandit problem with  $k = 5$  actions, denoted 1, 2, 3, 4, and 5. Consider applying to this problem a bandit algorithm using E-greedy action selection, sample-average action-value estimates, and initial estimates of  $Q_1(a) = 0$  for all  $a$ . Suppose the initial sequence of actions and rewards is  $A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 3, A_3 = 3, R_3 = 1, A_4 = 2, R_4 = 2, A_5 = 3, R_5 = 0, A_6 = 4, R_6 = 5$ . On some of these time steps the E case may have occurred causing an action to be selected at random. On which time steps did this definitely occur? On which time steps could this possibly have occurred?

#### References:

<https://smypybandits.github.io/notebooks/README.html>

<https://gist.github.com/pstthomas/4cf8b490a74ec1d7aec083b72c40c61a>

<https://www.kaggle.com/code/ilialar/multi-armed-bandit-vs-deterministic-agents>

<https://researchcode.com/code/1293156360/a-contextual-bandit-approach-to-personalized-news-article-recommendation/>

<https://towardsdatascience.com/solving-multi-armed-bandit-problems-53c73940244a>